GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

Pythagorean Theorem

Question Bank 6.03 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, find the hypotenuse. Give it exactly, and to the nearest tenth when it is not a whole number.

1. Legs of 8 cm and 6 cm

10 cm

2. Legs of 4 in and 11 in

√137 in ≈ 11.7 in

16 + 121 = 137.

3. Legs of 7 m and 8 m

√113 m ≈ 10.6 m

49 + 64 = 113.

For questions 4 through 6, find the missing leg. Give it exactly, and to the nearest tenth when it is not a whole number.

4. Hypotenuse 13 ft, one leg 12 ft

5 ft

169 − 144 = 25.

5. Hypotenuse 8 cm, one leg 5 cm

√39 cm ≈ 6.2 cm

64 − 25 = 39.

6. Hypotenuse 61 in, one leg 60 in

11 in

3,721 − 3,600 = 121 = 11^2.

7. A rectangle is 5 in wide and 12 in long. How long is its diagonal?

13 in

The diagonal is the hypotenuse of a 5-by-12 right triangle.

Choose one.

8. A right triangle has a hypotenuse of 25 m and one leg of 24 m. How long is the other leg?

- ○ A. 1 m
- ○ B. 34.7 m
- ● **C. 7 m**
- ○ D. 49 m

C

625 − 576 = 49 = 7^2.